

Assessment -8

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Question:

Demonstrate CRUD operations using SQLAlchemy.

Solution:

Aim:

To implement CRUD (Create, Read, Update, Delete) operations using SQLAlchemy ORM in Python.

Step 1: Install SQLAlchemy:

Bash

```
pip install sqlalchemy
```

Step 2: Create Python File (crud sqlalchemy.py):

Python

```
from sqlalchemy import create_engine, Column, Integer, String, Float
from sqlalchemy.orm import declarative_base, sessionmaker
```

```
# Database Connection
```

```
engine = create_engine('sqlite:///student.db', echo=True)
```

```
# Base Class
```

```
Base = declarative_base()
```

```
# Create Session
```

```
Session = sessionmaker(bind=engine)
```

```
session = Session()
```

```
# Model/Class Definition
```

```
class Student(Base):
```

```
    __tablename__ = 'students'
```

```
    id = Column(Integer, primary_key=True)
```

```
    name = Column(String(100))
```

```
    course = Column(String(100))
```

```
    marks = Column(Float)
```

```
    def __repr__(self):
```

```
        return f"Student(id={self.id}, name='{self.name}',  
course='{self.course}', marks={self.marks})"
```

```
# Create Table
```

```
Base.metadata.create_all(engine)
```

```
# ----- CREATE -----
```

```
def create_student():
```

```
    student = Student(  
        name="Nandini",  
        course="Python",  
        marks=95  
    )
```

```
session.add(student)
```

```
session.commit()
```

```
print("\nStudent Added Successfully")
```

```
# ----- READ -----
```

```
def read_students():
```

```
    students = session.query(Student).all()
```

```
    print("\nStudent Records:")
```

```
    for student in students:
```

```
        print(student)
```

```
# ----- UPDATE -----
```

```
def update_student(student_id):
```

```
    student = session.query(Student).filter(Student.id == student_id).first()
```

```
    if student:
```

```
        student.marks = 98
```

```
        session.commit()
```

```
        print("\nStudent Updated Successfully")
```

```
    else:
```

```
        print("\nStudent Not Found")
```

```
# ----- DELETE -----
```

```
def delete_student(student_id):
```

```
    student = session.query(Student).filter(Student.id == student_id).first()
```

```
    if student:
```

```
        session.delete(student)
```

```
        session.commit()
        print("\nStudent Deleted Successfully")
    else:
        print("\nStudent Not Found")
```

Execute CRUD Operations

```
create_student()
```

```
read_students()
```

```
update_student(1)
```

```
read_students()
```

```
delete_student(1)
```

```
read_students()
```

Close Session

```
session.close()
```

Output:

Student Added Successfully

Student Records:

```
Student(id=1, name='Nandini', course='Python', marks=95)
```

Student Updated Successfully

Student Records:

Student(id=1, name='Nandini', course='Python', marks=98)

Student Deleted Successfully

Student Records